Tight Ordered Width for Exponential-Threshold MCSP

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Abstract

Hardness magnification makes lower bounds for the Minimum Circuit Size Problem (MCSP) at exponential thresholds a natural test case. Let $A_{n,S} \subseteq \{0,1\}^N$, $N = 2^n$, be the set of truth tables computable by fan-in-two $\{\wedge, \vee, \neg\}$ circuits of size at most S. Fix $0 < \mu < 1$ and $S = \lfloor 2^{\mu n} \rfloor$. We show that for every permutation π of the coordinates, the minimum middle-layer width W of a strictly layered π-ordered branching program computing $A_{n,S}$ satisfies $\log_2 W = (1 + o(1))S \log_2 S$. This holds simultaneously for deterministic, nondeterministic and parity OBDDs and for field-weighted ordered algebraic branching programs, and the same asymptotic holds for minimum width and total size after optimizing over orders. The upper bound is the prefix trie of $A_{n,S}$, bounded by a tree-pin enumeration of circuits. For the lower bound, a packed-selector family has matching entropy; a Fourier averaging argument chooses a linear quotient whose fibres meet both halves of any balanced cut, so the family is recoverable from either side; and an equality-graph argument extracts an identity submatrix of order $2^{S \log S - O(S \log \log S)}$ in the communication matrix at the middle cut, which bounds the width via rectangle covers or field rank. The witnesses depend on the order. The result concerns ordered models only: it gives no lower bound for free or general branching programs, one-tape machines or circuits, and no consequence for *P* versus *NP*.

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Problem and contribution

Hardness magnification [1] shows that modest lower bounds for the Minimum Circuit Size Problem (MCSP) with threshold $S = 2^{\mu n}$ would have strong consequences, which makes exact lower bounds for MCSP in restricted models a natural test case. Existing branching-program lower bounds for MCSP [2,3] do not determine the complexity at thresholds $S = 2^{\mu n}$ with $0 < \mu < 1$. We determine the ordered-program complexity exactly at the logarithmic scale.

Main result (Theorem, Section 1): for fixed $0 < \mu < 1$, $S = \lfloor 2^{\mu n} \rfloor$ and every order π of the $N = 2^n$ truth-table coordinates, the minimum middle-layer width of a strictly layered π-ordered program computing MCSP_S satisfies $\log_2 W = (1 + o(1))S \log_2 S$, simultaneously for deterministic, nondeterministic, parity and field-weighted ordered programs.

The supplement contains a small check of Lemma 3: for twenty sampled balanced partitions at each of seven pairs (n, r) with $n \le 6$, it enumerates all row subspaces and verifies the minimum and average monochromatic-fiber inequalities in exact rational arithmetic. It does not exhaust all balanced partitions or compute Fourier coefficients, and it is not evidence for the asymptotic theorem.

Not claimed: no lower bound for free (non-oblivious) or general branching programs, one-tape machines or circuits, and no consequence for *P* versus *NP*.

1. Model and statement

All logarithms are base two. Circuits use binary AND and OR gates and unary NOT gates, with unbounded fan-out; the constants 0 and 1 may be used as inputs for free. The size of a circuit is its number of gates, including NOT gates; inputs are not counted. Let

$$\mathcal{L}_{m,s} = \{f : \{0,1\}^m \to \{0,1\} : \mathrm{CC}(f) \le s\}, \qquad a_m(s) = \log_2 |\mathcal{L}_{m,s}|.$$

For $N = 2^n$, identify an n-variable truth table with a vector in $\{0,1\}^N$, and write $\mathrm{MCSP}_S = A_{n,S} = \mathcal{L}_{n,S}$. A π-ordered program is strictly layered: every edge from layer $i-1$ to layer i reads coordinate $\pi(i)$, and every source–sink path

reads each coordinate once. For a program B, let $w_i(B)$ be its number of reachable states at layer i (unreachable states may be deleted), and define

$$W_i^{\pi,\mathsf{M}}(A) = \min\{w_i(B) : B \text{ is a strictly layered } \pi\text{-ordered program in model } \mathsf{M} \text{ computing } \mathbf{1}_A\}.$$

When the model and set are clear, we abbreviate this to W_i^{π}. A program's width is the maximum of its layer counts, and its total size is its number of vertices. A nondeterministic program accepts when a consistent path exists; a parity program uses the parity of the number of paths. In a field-weighted ordered algebraic program, every edge e has a fixed input-independent weight $w_e \in \mathbb{F}$ and a fixed bit label $b_e \in \{0,1\}$; on input x, only edges with $b_e = x_{\pi(i)}$ are retained at layer i, and the output is the source–sink path-sum over $\mathbb{F}$.

Main theorem

Fix $0 < \mu < 1$, let $S = \lfloor 2^{\mu n} \rfloor$, and let π be any permutation of the $N = 2^n$ coordinates. For each $\mathsf{M} \in \{\text{det}, \text{nondet}, \text{parity}, \text{field}(\mathbb{F})\}$, with $\mathbb{F}$ an arbitrary field,

$$S\log_2 S - C_\mu S \log_2\log_2 S \le \log_2 W_{N/2}^{\pi,\mathsf{M}}(\mathrm{MCSP}_S) \le S\log_2 S + C'_\mu S$$

for all $n \ge n_0(\mu)$, where the constants C_μ, C'_μ and $n_0(\mu)$ depend only on μ, not on π, the model or the field. In particular,

$$\boxed{\log_2 W_{N/2}^{\pi,\mathsf{M}}(\mathrm{MCSP}_S) = \Big(1 + O_\mu\Big(\frac{\log n}{n}\Big)\Big) S\log_2 S.} \tag{1}$$

The same bounds hold for the minimum width after optimizing over π, and for the minimum total size after enlarging C'_μ by 1 (Section 5). Uniformity of the error in π does not mean that one witness serves all orders; see (10).

2. Sharp semantic entropy

Proposition 1 (tree–pin enumeration)

For all $m, s \ge 2$,

$$a_m(s) \le (s+1)\log_2(m+s+2) + O(s). \tag{2}$$

In particular, if $s \ge m^2$, then $a_m(s) \le s\log_2 s + O(s)$.

Proof Prune a circuit to the gates reaching its unique output. For every non-output gate choose one actual consumer-pin occurrence. The chosen $q-1$ incidences form a rooted plane slot-tree (each node has at most two slots) directed toward the output; otherwise a directed cycle or a gate not reaching the output would remain. There are fewer than 4^q possible plane trees. If the circuit has b binary and u unary gates, then its $2b+u$ pins contain $q-1$ chosen tree pins, leaving

$$(2b+u) - (q-1) = b + 1 \le q + 1$$

free pins. The tree paths canonically name all gates. After the gate types are recorded, every free pin has at most $m+q+2$ possible sources (an input, a tree node, or one of two constants). Thus the number of codes with q gates is at most

$$4^q 3^q (m+q+2)^{q+1} \le 12^q (m+q+2)^{q+1}.$$

Summing over $q \le s$ proves (2). No division by $q!$ is used: a chain need not have many topological orders, while the selected consumer pins already provide the required encoding. □

Proposition 2 (packed-selector entropy lower bound)

There are constants C and s_0 such that, whenever $s \ge s_0$ and

$$m \ge \log_2 s + \log_2\log_2 s + C, \tag{3}$$

$$a_m(s) \ge s\log_2 s - O(s\log_2\log_2 s). \tag{4}$$

Proof

Let $L = \log_2 s$, $q = \lfloor L - 3\log_2 L \rfloor$, and let U_q be Lupanov's uniform upper bound for q-variable functions (Lupanov 1958, as presented in Theorem 1.15, Section 1.4, of [4]). That theorem is stated for fan-in-two circuits over the basis $\{\mathsf{AND}, \mathsf{OR}, \mathsf{NOT}\}$ whose size is the number of gates, NOT gates included, which is the convention of Section 1. It provides a bound U_q on the circuit size of every q-variable function with

$$U_q = \left(1 + O\left(\frac{\log q}{q}\right)\right)\frac{2^q}{q} + O(1).$$

Set $k = \lfloor s/(U_q + 7) \rfloor$ and $t = \lceil \log_2 k \rceil$. For every tuple of k arbitrary q-variable functions, a binary multiplexer on t address bits selects one component; unused leaves reuse the first component. Shared address negations make the multiplexer cost at most $3(2^t - 1) + t \le 7k$ gates. Restricting the address recovers the tuple, so the map is injective and the family has logarithmic cardinality $k2^q$. Since

$$q + t = \log_2 s + \log_2 \log_2 s + O(1), \qquad k2^q \ge s\log_2 s - O(s\log_2 \log_2 s),$$

(3) supplies enough variables and proves (4). □

For fixed $0 < \mu < 1$, $S = \lfloor 2^{\mu n} \rfloor$ eventually satisfies both (3) and $S \ge n^2$. Hence

$$a_n(S) = (1 + o(1))S\log_2 S. \tag{5}$$

3. A two-sided family for every balanced cut

Lemma 3 (Fourier quotient)

Let $P \sqcup Q = \mathbb{F}_2^n$ with $|P| = |Q| = 2^{n-1}$. For every $1 \le r < n$, there is a surjective linear map $L : \mathbb{F}_2^n \to \mathbb{F}_2^r$ such that the set

$$B_L = \{y \in \mathbb{F}_2^r : L^{-1}(y) \subseteq P \text{ or } L^{-1}(y) \subseteq Q\}$$

of monochromatic labels obeys

$$|B_L| \le 2^r \frac{2^r - 1}{2^n - 1}. \tag{6}$$

Proof

Put $\chi = 1_P - 1_Q$, and use normalized Fourier coefficients $\widehat{\chi}(\alpha) = \mathbb{E}_x \chi(x)(-1)^{\alpha \cdot x}$. For a codimension-$r$ subspace K, let

$$b_{x+K} = \mathbb{E}_{k \in K}\chi(x + k).$$

Conditional expectation and Parseval give

$$\mathbb{E}_{x+K} b_{x+K}^2 = \sum_{\alpha \in K^\perp} \widehat{\chi}(\alpha)^2.$$

A monochromatic coset contributes $1/2^r$ to the left side. Choose the r-dimensional space $K^\perp$ uniformly. Balance gives $\widehat{\chi}(0) = 0$, Parseval gives $\sum_{\alpha \ne 0} \widehat{\chi}(\alpha)^2 = 1$, and each nonzero character belongs to $K^\perp$ with probability $(2^r - 1)/(2^n - 1)$. Therefore some quotient satisfies (6). □

Lemma 4 (two-sided recoverable easy family)

For every balanced partition $P \sqcup Q$, there is a family $\mathcal{H}_{P,Q} \subseteq \mathcal{L}_{n,S}$ such that both restriction maps to P and to Q are injective and

$$\log_2 |\mathcal{H}_{P,Q}| \ge S\log_2 S - O(S\log_2 \log_2 S). \tag{7}$$

Proof

Assume n is large enough that $S_0 = S - 4n^2 > 0$, and apply the construction of Proposition 2 with budget S_0; it yields q, $k = \lfloor S_0/(U_q + 7) \rfloor$ and $t = \lceil \log_2 k \rceil$ with $k(U_q + 7) \le S_0$. Put $r = q + t = \log_2 S + \log_2 \log_2 S + O(1) < n$, so

$2^r = O(S \log S)$, and choose $L : \mathbb{F}_2^n \to \mathbb{F}_2^r$ by Lemma 3. Identify $\mathbb{F}_2^t$ with the addresses $0, \dots, 2^t - 1$, write $\mathbb{F}_2^r = \mathbb{F}_2^t \times \mathbb{F}_2^q$, and let

$$T = \{0, \dots, k-1\} \times \mathbb{F}_2^q, \qquad |T| = k2^q.$$

For $\theta : T \to \{0,1\}$ define $h_\theta(a,z) = \theta(a,z)$ for $0 \le a < k$ and $h_\theta(a,z) = \theta(0,z)$ for $k \le a < 2^t$; this is the packed selector applied to the components $\theta(a,\cdot)$, so $\mathrm{CC}(h_\theta) \le S_0$. Since $a \oplus b = (a \vee b) \wedge \neg(a \wedge b)$ uses four gates, the $r \le n$ linear forms of L cost at most $4r(n-1) < 4n^2$ gates, hence $\mathrm{CC}(h_\theta \circ L) \le S_0 + 4n^2 = S$. Set

$$\mathcal{H}_{P,Q} = \{h_\theta \circ L : \ \theta|_{T \cap B_L} = 0\} \subseteq \mathcal{L}_{n,S}.$$

For $y \in T \setminus B_L$ the fiber $L^{-1}(y)$ meets both P and Q, so either restriction of $h_\theta \circ L$ determines $\theta(y)$; both restriction maps are injective and $\log_2 |\mathcal{H}_{P,Q}| = |T \setminus B_L| \ge k2^q - |B_L|$. By Lemma 3, $|B_L| = O(2^{2r}/N) = O(S^2 \log^2 S/N)$, and as in Proposition 2 (with S_0 in place of S),

$$\log_2 |\mathcal{H}_{P,Q}| \ge S \log_2 S - O\Big(S \log_2 \log_2 S + n^2 \log_2 S + \frac{S^2 (\log_2 S)^2}{N}\Big).$$

For fixed $0 < \mu < 1$, $n^2 \log S = o(S \log \log S)$ and $S^2 (\log S)^2 / N = o(S)$, which gives (7). The constants depend only on μ, not on the partition. □

4. Identity extraction at the middle cut

Lemma 5 (equality graph)

Let $A \subseteq \{0,1\}^N$, let $P \sqcup Q = [N]$, and suppose $\mathcal{H} \subseteq A$ has both restrictions injective. Join distinct $f, g \in \mathcal{H}$ when either mixed table

$$f|_P \cup g|_Q \quad \text{or} \quad g|_P \cup f|_Q$$

lies in A. Then the graph has at most $|A|$ edges and has an independent set I with

$$|I| \ge \frac{|\mathcal{H}|^2}{2|A| + |\mathcal{H}|}. \tag{8}$$

For $f, g \in I$, the matrix

$$M_I(f,g) = 1_A(f|_P \cup g|_Q)$$

is the identity matrix.

Proof The map from ordered pairs (f,g) to the mixed table $f|_P \cup g|_Q$ is injective because both restriction maps are injective on $\mathcal{H}$. Thus at most $|A|$ oriented mixed tables can occur, and consequently the undirected graph has at most $|A|$ edges. The Caro–Wei (Turán) bound gives an independent set of size at least $|\mathcal{H}|^2/(2e + |\mathcal{H}|)$ in a graph with e edges, which is (8). Diagonal entries are in A; independence excludes both orientations of every off-diagonal pair, so the restriction matrix is identity. □

Since $\mathcal{H} \subseteq A$, we have $2|A| + |\mathcal{H}| \le 3|A|$, so (8) gives $\log_2 |I| \ge 2 \log_2 |\mathcal{H}| - \log_2 |A| - \log_2 3$. Taking $A = A_{n,S}$ and $\mathcal{H} = \mathcal{H}_{P,Q}$, Lemma 4 and Proposition 1 bound the right side by $2S \log_2 S - (S \log_2 S + O(S)) - O(S \log_2 \log_2 S)$, giving

$$\log_2 |I| \ge S \log_2 S - O(S \log_2 \log_2 S). \tag{9}$$

The witnesses have the order-dependent quantifiers

$$\forall \pi \ \exists L_\pi \ \exists \mathcal{H}_\pi \ \exists I_\pi. \tag{10}$$

They need not be efficiently found and are not one common witness for all orders.

5. Proof of the main theorem

Fix π and let P be its first $N/2$ coordinates and Q the remainder. Apply (9) to $A = A_{n,S}$. In a strictly layered π-NOBDD, choose an accepting path for each diagonal input indexed by I. Two such paths cannot share a middle state: combining the prefix of one with the suffix of the other would accept a forbidden off-diagonal table. Hence the middle width is at least $|I|$.

For a field-weighted ordered algebraic program, let $U_{f,z}$ be the sum of prefix-path weights from the source to middle state z on $f|_P$, and let $V_{z,g}$ be the corresponding suffix sum on $g|_Q$. Splitting paths at the middle layer gives

$$M_I = UV.$$

Since M_I is the identity over every field, $|I| = \operatorname{rank}(M_I) \le W_{N/2}$. Parity OBDDs are the case of $\mathbb{F}_2$, and deterministic OBDDs are the one-path special case. This proves the lower bound in (1).

For the upper bound, the prefix trie of the finite set $A_{n,S}$, in any order π, has at most $|A_{n,S}|+1$ states per layer. Proposition 1 gives

$$\log_2(|A_{n,S}| + 1) \le S \log_2 S + O(S).$$

The trie has at most $N(|A_{n,S}| + 1)$ nodes. For fixed $\mu > 0$ we have $\log_2 N = n \le \mu^{-1} \log_2(S + 1) = o(S)$, so the extra factor N is absorbed into $O_\mu(S)$: the minimum total size satisfies the upper bound in (1) with C'_μ replaced by $C'_\mu + 1$ for all sufficiently large n. Since total size is at least width, the lower bound transfers unchanged. Thus width and total size satisfy the same explicit bounds, completing the proof. □

6. Scope and literature boundary

The theorem applies to strictly layered ordered programs. It does not apply to free or path-dependent FBDDs/1-NBPs, co-nondeterministic programs, general branching programs, bounded-error randomized programs, or one-tape time. The order-dependent witness quantifier in (10) is essential; there is no claim of a single identity submatrix working for every order.

Cheraghchi, Hirahara, Myrisiotis and Yoshida [2] prove that read-once co-nondeterministic branching programs computing MCSP need size $2^{\tilde{\Omega}(N)}$ [2, Corollary 7, STACS version], and remark [2, Section 1.2.1] that their argument implicitly gives an exponential lower bound for read-once oblivious branching programs; deterministic π-OBDDs are a special case of both models. Their parameter regime and methods (hitting-set generators) differ from ours; their exponential bound for ordered programs concerns deterministic read-once oblivious programs and comes without a matching upper bound. For MCSP, the $N^{1.5-o(1)}$ size lower bound of [2, Theorem 5, STACS version] covers nondeterministic, co-nondeterministic and parity branching programs and relies on the Impagliazzo–Meka–Zuckerman pseudorandom generator; such bounds apply to ordered subclasses as well. The Nečiporuk argument in [2, Theorem 4] instead establishes lower bounds for MKTP. Our comparison concerns only exponential bounds for ordered programs: there we did not find in [2] a matching exponent, or exponential bounds for nondeterministic, parity or field-weighted ordered programs. Glinskih and Riazanov [3] prove an $N^{\Omega(\log \log N)}$ lower bound in the stronger, path-dependent model of read-once nondeterministic branching programs. Hardness magnification [1] and related circuit lower bounds for MCSP [5] motivate the one-tape question discussed below. Volk [6] recently proved an almost-quadratic lower bound for an explicit function against unrestricted read-once parity branching programs; that model is stronger than the ordered parity programs considered here, and [6] does not address MCSP. Our contribution relative to these works is the matching exponent $S \log_2 S$, with explicit error terms, for every fixed order and an identity-submatrix witness that applies uniformly to four ordered models. We do not claim that this combination is absent from all of the literature; it is what we found in the works cited here.

The present argument establishes lower bounds for ordered programs only. It does not establish the one-tape running-time lower bounds required by the hardness-magnification results cited above, and no complexity-class separation such as $P \neq NP$ or SAT $\notin P/\mathrm{poly}$ is claimed.

References

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